

MIT Academy of Engineering, Alandi(D), Pune

Department of CSE AIML

DV MINI PROJECT

Course Code: 2311162L

Course Name: Data Visualization Lab

Academic Year: 2025–26

Student Details

- **Student Name:** Ashish Kalaskar and Aryan Bhosale
- **PRN Number:** 202501110176 and 202501110179
- **Division / Batch:** AI2 A23
- **Dataset Name & Source:**
- **Description:**

The *Student Placement Dataset* belongs to the education and employability domain and contains information about students' academic performance, skills, and placement outcomes. It is used to analyze how factors like marks, specialization, and work experience influence job placement and salary. The dataset helps in identifying patterns that can improve student career readiness and institutional training strategies.

<https://www.kaggle.com/datasets/sonalshinde123/student-placement-dataset>

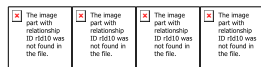
Dashboard link: https://mitaoepune-my.sharepoint.com/:u:/g/personal/202501110176_msteams_mitaoe_ac_in/IQCHhOy5fcWQS7eUAK4Tu9aMAfQEPjYKHEns9jqAd7k-AOQ?e=EiMbmV

Title: Design of Interactive Dashboards and Storyboards for Analytical Storytelling

Problem Statement

Construct an interactive dashboard and storyboard using best visualization, formatting, and storytelling practices to communicate insights effectively.

1. Data Loading



Power BI Desktop interface showing the 'test.csv' data source being imported. The 'Data Type Detection' window is open, displaying a table of student data with columns: Student_ID, Age, Gender, Degree, Branch, CGPA, Internships, Projects, Coding_Skills, Communication_Skills, and Aptitude_Te.

Student_ID	Age	Gender	Degree	Branch	CGPA	Internships	Projects	Coding_Skills	Communication_Skills	Aptitude_Te
15202	23	Male	B.Tech	Civil	7.32	2	4	5	6	6
4573	24	Female	MCA	ME	4.76	0	1	1	1	4
34424	20	Male	BCA	ME	6.16	0	3	3	3	8
38881	19	Male	B.Sc	IT	8.77	2	5	8	5	5
30191	23	Male	B.Tech	ME	7.63	0	3	4	4	6
13635	21	Female	B.Tech	IT	7.11	1	3	4	3	3
24929	20	Male	MCA	IT	7.66	2	5	8	4	4
32313	19	Male	B.Sc	IT	7.08	2	5	8	5	5
7207	23	Male	B.Tech	IT	7.1	2	5	9	4	4
6914	18	Female	BCA	ECE	7.65	0	3	4	4	5
27120	21	Female	B.Sc	IT	8.66	0	4	6	4	4
13254	21	Male	B.Sc	ME	7.04	1	3	4	4	4
49098	20	Male	B.Tech	CSE	7.07	1	5	8	5	5
47647	22	Male	BCA	CSE	7.49	3	5	8	3	3
43993	23	Male	B.Sc	CSE	5.92	1	4	7	4	4
5880	21	Female	BCA	IT	7.94	2	5	8	4	5
25053	22	Female	B.Tech	ME	7.77	0	3	4	4	5
42721	18	Female	B.Sc	ME	6.58	0	3	4	4	6
16636	18	Male	B.Sc	ME	7.58	0	4	5	5	5
41887	19	Male	B.Tech	CSE	5.55	0	4	8	6	6

The 'Visualizations' pane on the right shows the 'Build visual' button and a search bar. The 'Data' pane shows the 'test.csv' data source.

2. Data Transformation (cleaning, preprocessing steps)

Power Query Editor interface showing the 'Table.TransformColumnTypes' step. The 'Column statistics' pane is open, displaying the distribution of data for each column. The 'Value distribution' chart shows the frequency of values for each column.

The 'Column statistics' pane shows the following data:

Column	Count	Error	Empty	Distinct	Unique	NaN
Student_ID	1000	0	0	368	118	0
Age	1000	0	0	7	7	0
Gender	1000	0	0	2	2	0
Degree	1000	0	0	4	4	0
Branch	1000	0	0	5	5	0
CGPA	1000	0	0	368	368	0
Projects	1000	0	0	6	6	0
Coding_Skills	1000	0	0	10	10	0

The 'Value distribution' chart shows the frequency of values for each column. The 'Column statistics' pane also shows the 'Valid', 'Error', and 'Empty' counts for each column.

The 'Table.TransformColumnTypes' step is applied to the 'test' query, with the following parameters: (*Promoted Headers), ({Student_ID, Int64.Type), (Age, Int64.Type), (Gender, type text), (Degree, type text), (Branch, type text), (CGPA, type number), (Projects, type number), (Coding_Skills, type number).

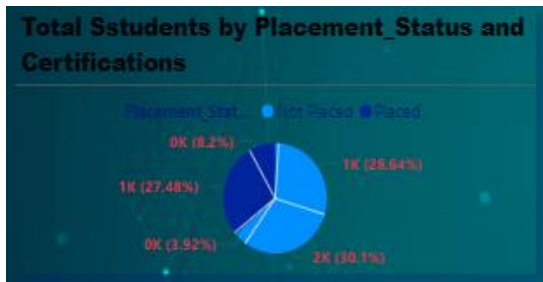
The 'Query Settings' pane shows the 'Properties' and 'Applied Steps' for the query. The 'Properties' pane shows the 'Name' and 'Source' of the query. The 'Applied Steps' pane shows the 'Changed Type' step.

6. Grain (Level of Data Detail)

the grain of the dataset is defined as **one row per student's placement profile**.

Each record represents a single student containing academic details (such as CGPA and degree), skill-related attributes (such as coding and communication skills), experience-related data (such as internships and certifications), and the final placement outcome (placed or not placed).

7 Selection of Appropriate Visuals .



This visual helps quickly compare **placed vs non-placed students** based on certification status in a single view.

It highlights whether certifications are improving placement outcomes and shows percentage distribution clearly.

Useful for identifying skill gaps and making better training/placement decisions.

Insight:

A higher percentage of certified students appear to be placed compared to non-certified students, indicating certifications positively influence placement success. Students without certifications may require additional upskilling programs



Branch-wise Communication Skill Distribution Analysis

Description:

This visual compares communication skill levels across different branches (IT, CSE, ECE, ME, Civil) to identify which departments have stronger or weaker communication abilities. It helps organizations detect soft-skill gaps, plan targeted training programs, and improve student placement readiness

Branch-wise Coding Skill Distribution Analysis

Some branches show stronger communication skill scores than others, highlighting uneven soft-skill development across departments. Branches with lower communication scores may need focused training for interview readiness.

This visual shows coding skill levels across different branches to compare technical proficiency among students. It helps identify departments with strong coding capabilities, detect skill gaps, and design targeted training programs to improve placement outcomes

Technical branches such as IT/CSE likely show stronger coding performance compared to non-core branches. Lower-performing branches may benefit from additional coding bootcamps and technical workshops..



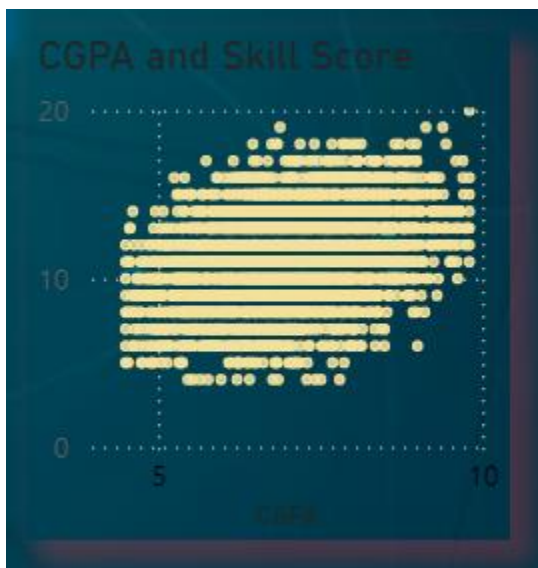
Skill Score vs Placement Outcome Analysis

Description:

This visual illustrates the relationship between students' skill scores and their placement status (placed vs not placed). It helps identify the skill score range where placements are highest, enabling institutions to set performance benchmarks and improve student readiness.



CGPA vs Skill Score Correlation Analysis



Description:

This scatter plot shows the relationship between students' academic performance (CGPA) and skill scores. It helps identify whether higher academic scores align with stronger practical skills and supports better placement and training decisions.

Insight:

Students with higher overall skill scores have a greater probability of getting placed. Placement rates decline significantly for students with lower skill scores, suggesting a minimum skill threshold for employability.

8. Create Minimum 5 Measures (DAX)

Total Students = COUNTROWS('Fact_Placement')

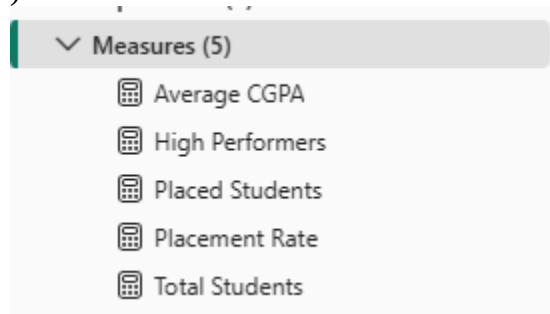
Placed Students = CALCULATE(COUNTROWS('Fact_Placement'),
'Fact_Placement'[Placement_Status] = "Placed")

Placement Rate = DIVIDE([Placed Students], [Total Students])

Average CGPA = AVERAGE('Fact_Placement'[CGPA])

High Performers =
CALCULATE(
COUNTROWS(Fact_Placement),
Fact_Placement[CGPA] >= 8,
Fact_Placement[Placement_Status] = "Placed"

)



9. Create minimum 5 Calculated Columns

CGPA Category =

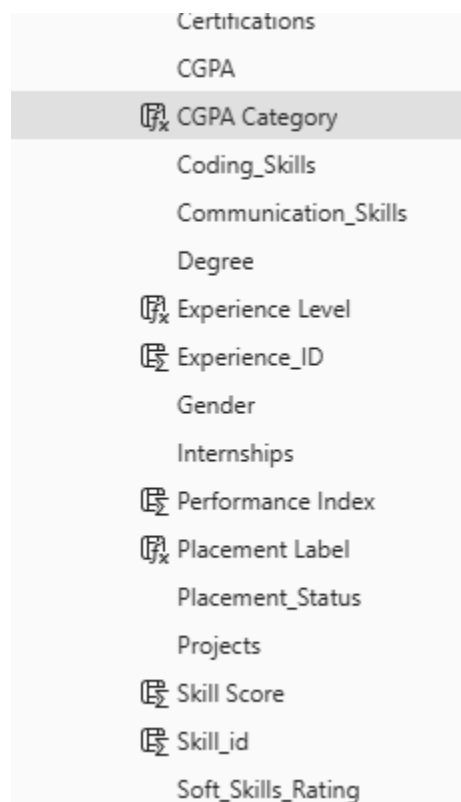
```
IF(
    Fact_Placement[CGPA] >= 8, "High",
    IF(Fact_Placement[CGPA] >= 6, "Medium", "Low")
) Placement Label =
IF(Fact_Placement[Placement_Status] = "Placed", "Yes", "No") Skill Score =
Fact_Placement[Coding_Skills] +
```

Fact_Placement[Communication_Skills] Performance Index =

Fact_Placement[CGPA] * Fact_Placement[Aptitude_Test_Score] Experience Level =
IF(Fact_Placement[Internships] >= 2, "High", "Low")



```
1 CGPA Category =
2 IF(
3     Fact_Placement[CGPA] >= 8, "High",
4     IF(Fact_Placement[CGPA] >= 6, "Medium", "Low")
5 )
```



Certifications
CGPA
CGPA Category
Coding_Skills
Communication_Skills
Degree
Experience Level
Experience_ID
Gender
Internships
Performance Index
Placement Label
Placement_Status
Projects
Skill Score
Skill_id
Soft_Skills_Rating

11. Publish Report to Power BI Service

The Power BI report was published to the Power BI Service using the Publish option available in Power BI Desktop. After signing in, the report was uploaded to a selected workspace where both the dataset and report became available online. This allows the dashboard to be accessed, shared, and refreshed through the cloud-based Power BI Service

12. Map your project to a relevant SDG Goal

SDG 4: Quality Education

✓ Why it fits:

- Analyzes student performance
- Identifies skill gaps
- Improves learning outcomes
- Helps students become job-ready

SDG 8: Decent Work & Economic Growth

✓ Why it fits:

- Predicts placement chances
- Analyzes employability factors
- Helps improve job readiness
- Links education to employment
-

3. Creating Data Model

New relationship

Select tables and columns that are related.

From table

Fact_placement

	Internships	Placement_St...	Projects	Skill_id	Soft_Skills_Ra...	Student_ID
	0	Placed	4	null	6	12101
0		Not Placed	4	null	6	10520
0		Not Placed	4	null	6	19133

To table

Fact_Table

Aptitude_Test...	CGPA	Placement_St...	Student_ID
72	8.03	Placed	12101
51	6.71	Not Placed	10520
86	7.83	Not Placed	19133

Cardinality

One to many (1:*)

Cross-filter direction

Both

☐ Make this relationship active

☐ Apply security filter in both directions

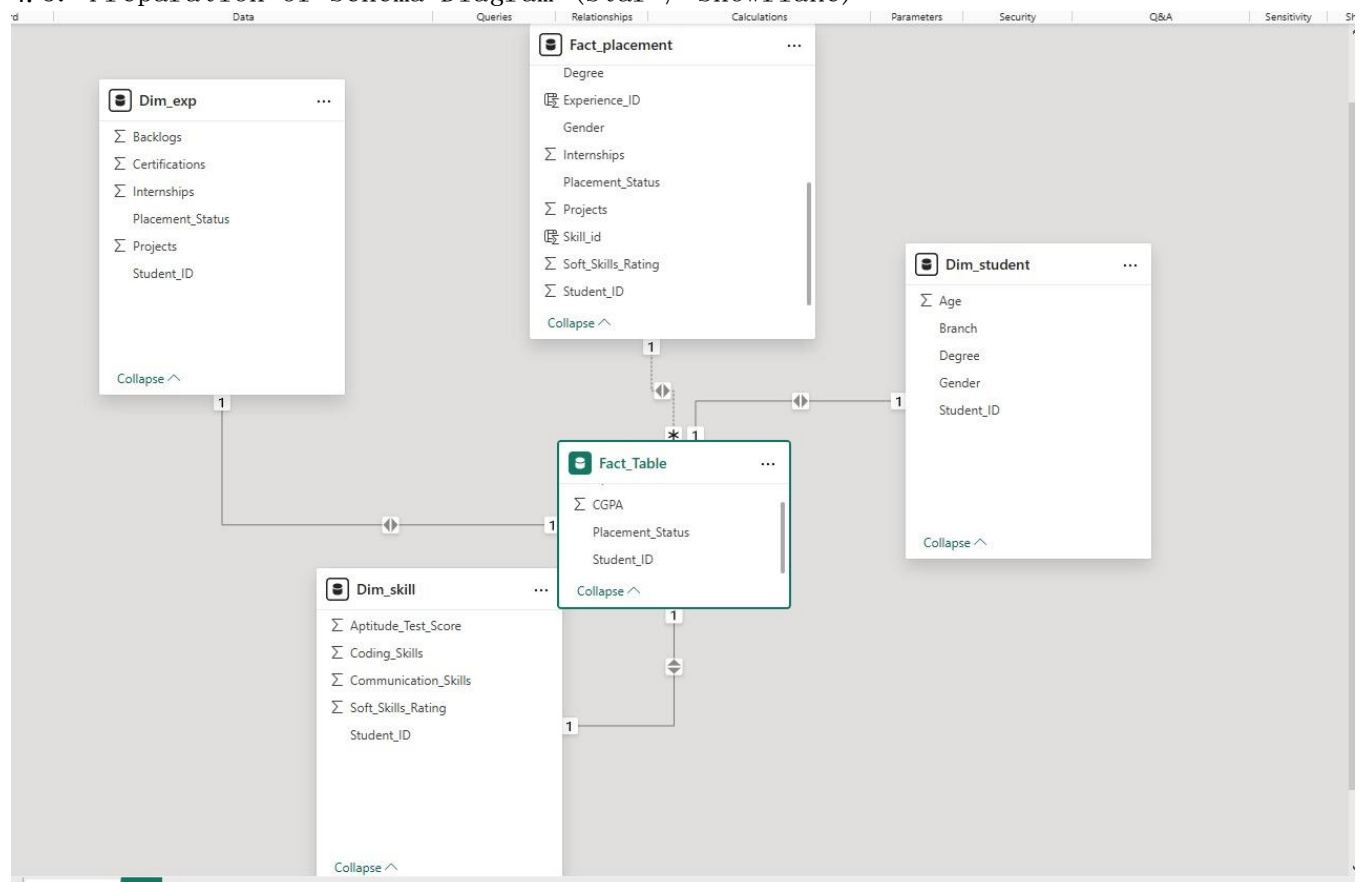
☐ Assume referential integrity

Save

Cancel

3.

4.5. Preparation of Schema Diagram (Star / Snowflake)



FINAL DASHBOARD

